

Exploring Written Artefacts in Virtual Reality – Potential and Challenges for Creating Immersive Tools and Applications

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Figure 1: Our interactive VR applications with markers indicating the location of all Written Artefacts. Left: scenario (a) ancient theatre of Miletus, right: interior of the church in Lucklum

Abstract

In this work, we introduce two interactive virtual reality (VR) applications, so-called Immersive Incribed Spaces, for researching Written Artefacts in our interdisciplinary projects. We discuss the potential of using VR to access historical cultural heritage sites and how the immersive experience facilitates individual perception of Written Artefacts. For this novel context, we explore potential use cases and present our implemented features for the VR applications. Furthermore, we examine two major challenges: (1) Spatial interaction and (2) visualization of spatial information. Finally, we describe how we integrated improvements into our applications, based on our research results.

Index Terms: Human-computer interaction, classical archaeology, ancient history, epigraphy, theology, inscriptions, cultural heritage, virtual reality, interactive systems

1 Introduction

Modern standalone head-mounted displays (HMDs) greatly facilitate the setup and usage of HMDs for a wider audience and in new areas of research. One of which is

the research on so-called *Written Artefacts* – a cover term for natural or artificial objects bearing visual signs applied or formed by humans, regardless of their material composition (materiality). Written Artefacts are an essential part of our collective cultural heritage and include, among others, written manuscripts, inscriptions, pictorials (drawings/illustrations), and ancient graffiti [1, 15]. In our previous work, we applied Virtual Reality (VR) technology for research in two interdisciplinary projects involving researchers from classic archaeology, ancient history, theology, literary and art history, and human-computer interaction [5, 6]. Both projects were carried out within the Cluster of Excellence "Understanding Written Artefacts" at the University of Hamburg. We created realistic, true-to-scale 3D VR representations with high-resolution textures of two real-world locations with Written Artefacts. These *Immersive Incribed Spaces* spaces can be freely explored and their Written Artefacts including additional information can be viewed in their spatial context. We also added 3D reconstructions of former structures and developed various features which encourage users to gain new perspectives and raise new research questions on the Written Artefacts and their spatial constructions. Within our interdisciplinary collaboration, we defined use cases and explored innovative use cases for VR in the context of researching Written Artefacts that we will discuss in this paper. However, these new application scenarios come with their specific requirements and issues for interaction in VR. During our work, we identified two major interaction challenges for our *Immersive Incribed Spaces* that we addressed and investigated in several user studies. This paper presents how we integrated the findings into our *Immersive Incribed Spaces*.

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2 Immersive Inscribed Spaces

In our projects we worked on two inscribed spaces with contrasting properties as pictured in Figure 1:

- (a) The ancient Greco(Greek)-Roman theatre of Miletus (Turkey). It is one of the largest preserved ancient theatres, which is a large-scale open space with over 300 (Ancient Greek) Written Artefacts carved in marble stone.
- (b) The interior of the early modern European church in Lucklum (Germany) is an enclosed interior space with 209 Latin inscriptions and 156 emblematic paintings (monochromatic paintings using the European *Grisaille* technique) representing biblical themes.

To summarise, both spaces contain a large amount of Written Artefacts from entirely different time periods and with different materiality. Regardless of these differences, both scenarios can benefit equally from the immersive experience of the space and the Written Artefacts. The use of VR for investigating Written Artefacts is independent of the geography, materiality, time period, and architecture of the inscribed space [2]. The *Immersive Incribed Spaces* that we created allow the humanities researchers in our projects to investigate diverse, multifaceted questions. They support the researchers in gaining historical insights and investigating new questions based on the immersive, spatial perception of the Written Artefacts that are inaccessible in the real-world locations. Our previous evaluations showed that usability and sense of spatial presence were generally rated high [6, 8]. We will describe the core features and benefits of the *Immersive Incribed Spaces* in the following subsections. Further details on the user-centred design and development process and the implemented features can be found in our previous work [6, 8].

2.1 Providing Spatial Context and Restoring Visibility

A key question for the humanities researchers is how people used to structure and interact with a space, and how they occupied and represented themselves in it. The *Immersive Incribed Spaces* can display additional temporal layers, allowing researchers to view former conditions that no longer exist, e.g. due to decay of structures in scenario (a), as shown in Figure 2 or overpainting in scenario (b). For scenario (a), the 3D reconstructions of former structures visualise multiple time periods of the ancient theatre. Moreover, they provide a non-invasive alternative to anastylosis, the re-erection of non-preserved architectural elements in the real world. With anastylosis, only one time period can be restored, and the process often results in damage to ancient structures. Combined with a time of day simulation, the reconstructions reveal how much shadow was cast onto inscriptions. In scenario (b), virtual overlays of non-invasive X-ray Fluorescence (XRF)-scans uncover hidden layers in an emblematic painting.

In addition, the 3D VR representations of both cultural heritage sites preserve their current state in case of further destruction or decay.

2.2 Gaining New Perspectives

Another important aspect is the ability to view inaccessible locations and engage with new or impossible perspectives of these spaces in VR. We implemented various overview spots, including a bird's-eye perspective for scenario (a) that offers a complete top down overview of the ancient theatre and a close-up view of Written Artefacts on the ceiling of the church for scenario (b).



Figure 2: 3D reconstructions of former structures of the ancient theatre, including the stage building and seating tiers.

2.3 Visualising and Filtering Written Artefacts

We added spatial markers that provide an overview of the distribution and clustering of all the Written Artefacts in their inscribed spaces. These markers can be selected to open up spatial information panels for each Written Artefact. These markers can be displayed in a filtered view based on their properties. For scenario (a) the markers indicate different groups of people (e.g. Jews), time periods, and functions of the Written Artefacts as displayed in Figure 3. Colour coding was used to distinguish between the different types, consisting of inscriptions, pictorial elements, game boards, and architectural marks. Filtering was used in scenario (b), as seen in Figure 4, to visualise the distribution of different themes and religious iconography of the Written Artefacts within the church.

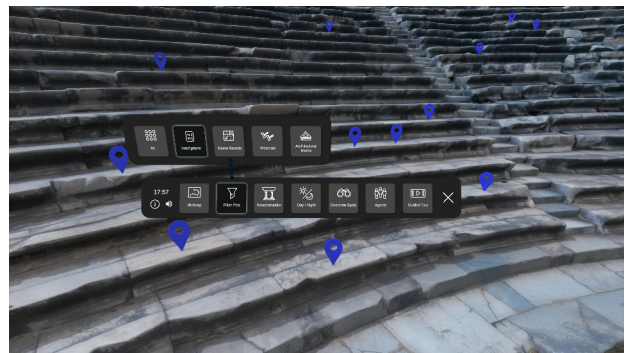


Figure 3: Menu for scenario (a) showing a filtered selection of Written Artefacts of the type *inscription*.

2.4 Reference, Information, Communication

Due to the large amount of Written Artefacts, the spatial information panels with their numbering are important for providing reference to the researchers. However, they are not only used as reference guides by researchers. They also contain proper, easier to read transcriptions and translations of the Written Artefacts, which makes them more accessible for other researchers, students, or interested users in general. The illustrations and photographs on the panels offer different perspectives and zoomed in detail views of Written Artefacts and the commentary text explains their social, cultural, and historical context.



Figure 4: Menu for scenario (b) with a filtered view of based on the selected iconographic category.

3 Interaction Challenges in VR

We identified two major challenges for interacting with our *Immersive Incribed Spaces* in VR that we studied in our research:

1. **Spatial Interaction – Interacting with Written Artefacts:** The large amount of Written Artefacts are distributed throughout the *Immersive Incribed Spaces*. Their 3D position markers form clusters and overlap. This makes precise and distant spatial selections difficult.
2. **Spatial Information – Presenting Written Artefacts Information:** As detailed explanations, texts, illustrations, and photographs are integrated in VR, it's challenging to present the amount of content in a spatial context.

We explored and implemented possible solutions for both challenges and evaluated them in multiple user studies. The results and findings were integrated into our *Immersive Incribed Spaces*.

3.1 Interacting with Written Artefacts

Raycasting is one of the most frequently used spatial interaction techniques in modern HMDs. It is easy to understand and use, as it is based on the pointer metaphor. However, selection performance significantly deteriorates with decreasing target size and increasing target distance [14, 13]. Our *Immersive Incribed Spaces* require flexible selection of close as well as distant Written Artefacts markers that are clustered or close together. We proposed and evaluated redirected raycasting for facilitating spatial selection interaction [9, 10]. This assistive technique uses gain-based movement interpolation, similar to the already established redirected walking technique for VR locomotion, to subtly redirect the raycast towards the closest target. When the raycast enters a specified redirection zone, it is gradually pulled towards the redirection centre, which counters movement and tracking jitter to facilitate and stabilise the selection. We developed and evaluated three different raycast visualizations with different degrees of ray curvature and a gaze-assisted redirection technique. The results of our two user studies suggest that raycast redirection significantly improves selection performance and decreases perceived workload compared to classic raycasting (Baseline). Different visualizations did not affect the performance, but they might

influence the perception of the redirection. In both studies, redirection did not negatively affect the sense of agency of the users. Based on these promising findings, we implemented raycast redirection into our application for scenario (a), the Ancient Theatre of Miletus, as pictured in Figure 5. This large open space with scattered and clustered Written Artefacts makes raycast selection of the 3D markers especially challenging. With the improved assistive raycasting, we could also increase the maximum ray interaction distance to 20 meters in VR.



Figure 5: Illustration of visualized actual tracked raycast position (right ray) and the redirected ray pointer (left ray) shown to the user.

3.2 Presenting Written Artefacts Information

For our specific use case involving Written Artefacts, it is essential to include various texts and descriptions in our spatial information panels. This requires users, especially for scenario (b) the interior of the church in Lucklum, to read long texts in VR. Although the increased display resolution of modern HMDs makes reading long texts feasible, there are still many challenges for visualizing textual information that have not been examined in detail. While there are some studies on text colours and text parameters such as font size [3, 4, 11, 12] there are no insights on layouts and user interface (UI) controls for accessing and reading long texts in VR. Thus, our previous work focused on performance and UX aspects for presenting and reading long texts in VR [7]. We developed four different text panels based on familiar desktop UI patterns, two discrete variants using pagination and two continuous variants with a scrollbar only, and scrollbar and scroll-buttons. Our user study revealed that the different panels did not influence the recall of content. This indicates that established desktop UI patterns can be applied to spatial UI. While some users liked using buttons, others preferred navigating with the scrollbar. However, the scrollbar only variant was the least preferred. Therefore, we adjusted our spatial information panels for scenario (b) and used the combination of scrollbar and scroll-buttons as UI controls for the long description texts as seen in Figure 6a. Since a lot of different information as well as images and photos need to be displayed, we created an adaptive spatial panel design based on input and feedback from the humanities researchers. As pictured in Figure 6b, the panel is set up in a default, tabbed mode that contains different pages but can be switched to an alternative, side-by-side view with all pages arranged as a curved display to allow individual access for reading and comparing information.



(a) Tabbed view

(b) Side-by-side view

Figure 6: Spatial information panel in default tabbed view mode (left) and alternative side-by-side mode (right).

4 Conclusion

Through our interdisciplinary work on *Immersive Inscribed Spaces* we identified and implemented various features that demonstrate the potential of VR for research on Written Artefacts: It gives researchers access to impossible perspectives, provides them with a systematic overview of large amounts of Written Artefacts, and unlocks temporal context by restoring visibility of hidden layers and former, non-preserved structures. VR enables the immersive experience of historical cultural heritage sites and facilitates individual perception of Written Artefacts in their spatial context. However, we also showed that the use of immersive technologies in the new context for researching Written Artefacts brings specific, new challenges for the XR community regarding technical implementation, interaction techniques, and UI visualizations. We believe that interdisciplinary collaboration will be crucial for addressing these issues and for creating purposeful XR-based tools and applications for cultural heritage. We believe that these novel tools and applications will not only benefit researchers, but could also be used in science communication and for raising public awareness of the importance of cultural heritage and the research field of Written Artefacts.

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References

- [1] A. Bausi, D. Bondarev, N. Bregler, S. Chiarini, G. Ciotti, J. Droese, E. Dal Sasso, M. Friedrich, A. Helman-Wažny, M. Kohs, L. Mascia, A. L. Osthof, M. Ratzke, O. Škrabal, S. Sövegjártó, T. Staack, and H. Wimmer. Definition of ‘written artefact’. This paper is an occasional paper. Its results are based on exchanges, reflections, and discussions of the Theory and Terminology working group., Nov. 2023. doi: 10.25592/uhhfdm.13836 1
- [2] C. Berns and A. Osthof. *Streets as contested spaces in ancient Miletus*, pp. 131–141. 2024. 2
- [3] T. Dingler, K. Kunze, and B. Outram. Vr reading uis: Assessing text parameters for reading in vr. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, CHI EA ’18, p. 1–6. Association for

- Computing Machinery, New York, NY, USA, 2018. doi: 10.1145/3170427.3188695 3
- [4] A. Erickson, K. Kim, G. Bruder, and G. F. Welch. Effects of dark mode graphics on visual acuity and fatigue with virtual reality head-mounted displays. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 434–442. IEEE, Manhattan, NY, USA, 2020. doi: 10.1109/VR46266.2020.00064 3
- [5] J. Gabel, C. Berns, S. Bosch, J. Eickmeyer, K. Harter-Uibopuu, N. Martin, A. L. Osthof, J. A. Steiger, and F. Steinicke. User-Centered Design of Immersive Research Applications for Understanding Written Artefacts. In J. Orlosky, D. Reiners, and B. Weyers, eds., *ICAT-EGVE 2021 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments*. The Eurographics Association, 2021. doi: 10.2312/egve.20211325 1
- [6] J. Gabel, C. Berns, S. Bosch, J. Eickmeyer, K. Harter-Uibopuu, N. Martin, A. L. Osthof, J. A. Steiger, and F. Steinicke. Immersive inscribed spaces – bringing virtuality to written artefacts for humanities. *i-com*, 21(1):7–18, 2022. doi: doi:10.1515/icom-2022-0012 1, 2
- [7] J. Gabel, M. Ludwig, and F. Steinicke. Immersive reading: Comparison of performance and user experience for reading long texts in virtual reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI EA ’23. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3544549.3585895 3
- [8] J. Gabel, S. Rings, and F. Steinicke. Ancient theatre of miletus: A virtual reality research tool for studying written artefacts. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction, SUI ’23*. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3607822.3618021 2
- [9] J. Gabel, S. Schmidt, O. Ariza, and F. Steinicke. Redirecting rays: Evaluation of assistive raycasting techniques in virtual reality. In *Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology, VRST ’23*. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3611659.3615716 3
- [10] J. Gabel, S. Schmidt, K. Pfeuffer, and F. Steinicke. Guiding handrays in virtual reality: Comparison of gaze- and object-based assistive raycast redirection. In *Proceedings of the 2024 ACM Symposium on Spatial User Interaction, SUI ’24*. Association for Computing Machinery, New York, NY, USA, 2024. doi: 10.1145/3677386.3682080 3
- [11] S. Kobayashi, K. Kanari, and M. Sato. An examination of view-settings for long texts in vr reading. In *ACM SIGGRAPH 2021 Posters*, SIGGRAPH ’21. Association for Computing Machinery, New York, NY, USA, 2021. doi: 10.1145/3450618.3469164 3
- [12] T. Kojić, D. Ali, R. Greinacher, S. Möller, and J.-N. Voigt-Antons. User experience of reading in virtual reality — finding values for text distance, size and contrast. In *2020 Twelfth International Conference on Quality of Multimedia Experience (QoMEX)*, pp. 1–6. IEEE, Manhattan, NY, USA, 2020. doi: 10.1109/QoMEX48832.2020.9123091 3
- [13] J. J. LaViola, E. Kruijff, R. P. McMahan, D. A. Bowman, and I. Poupyrev. *3D User Interfaces - Theory and Practice*. Addison-Wesley, Redwood City, CAUnited States, 2nd ed., 2017. 3
- [14] I. Poupyrev, T. Ichikawa, S. Weghorst, and M. Billinghurst. Egocentric Object Manipulation in Virtual Environments: Empirical Evaluation of Interaction Techniques. *Computer Graphics Forum*, Volume17(Issue3):41–52, 1998. doi: 10.1111/1467-8659.00252 3
- [15] O. Škrabal, L. Mascia, A. Osthof, and M. Ratzke, eds. *Graffiti Scratched, Scrawled, Sprayed: Towards a Cross-Cultural Understanding*. De Gruyter, Berlin, Boston, 2023. doi: doi: 10.1515/9783111326306 1